

Medical Image Processing Pipeline for Cancer CellDetection

CEP-415: Digital Image Processing Lab

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Introduction

System Architecture

Results

Mathematical Foundation

Sustainable Development Goals

Future Enhancements

Conclusions

Introduction

Problem Statement

- ▶ **Challenge:** Early cancer detection is crucial for patient survival
- ▶ Medical image analysis is time-consuming and requires expert radiologists
- ▶ Manual analysis is prone to human error and variability
- ▶ Limited access to specialised expertise in resource-constrained regions
- ▶ **Solution:** Automated image processing pipeline for objective feature extraction

Objective: Develop a robust, multi-stage image processing system for automated medical image analysis

Key Objectives

Primary Goals:

- ▶ Enhance diagnostic image contrast
- ▶ Extract structural features
- ▶ Identify tumour boundaries
- ▶ Reduce noise and artefacts
- ▶ Enable multi-aspect analysis

Technical Requirements:

- ▶ Scalable microservices architecture
- ▶ Real-time processing capability
- ▶ Cloud infrastructure integration
- ▶ Quantitative quality metrics
- ▶ Clinical validation support

Accuracy

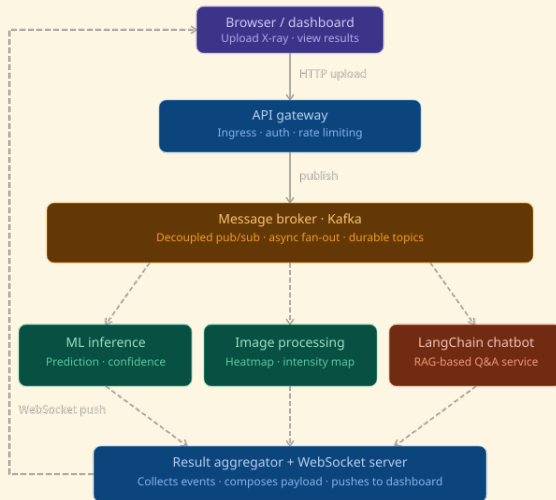
Speed

Equity

Scalability

System Architecture

System Architecture Overview



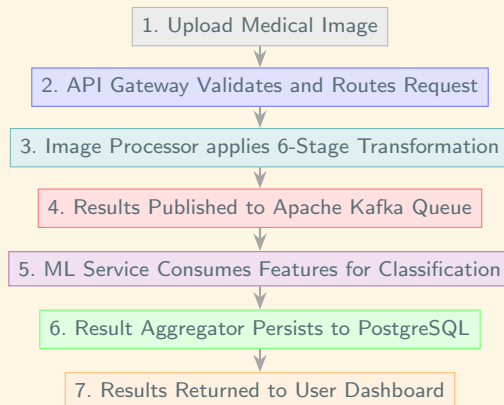
→ Sync (HTTP)

- - - - -> Async event (pub/sub)

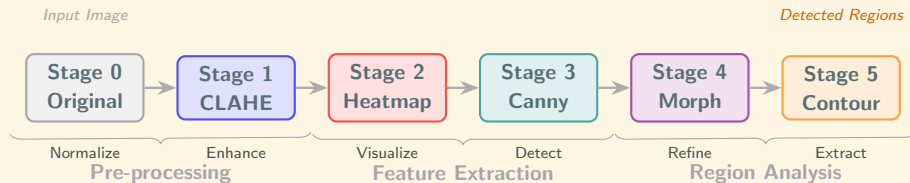
- - - - -> WebSocket return

Cancer Cell Detection Pipeline

Data Processing Flow



6-Stage Image Processing Pipeline



Stage 0: Original Image

Grayscale Normalisation:

- ▶ Load image as 8-bit grayscale (0–255)
- ▶ Normalise: $I_{norm} = \frac{I_{original}}{255}$
- ▶ Establish baseline for subsequent stages
- ▶ Compute intensity histogram

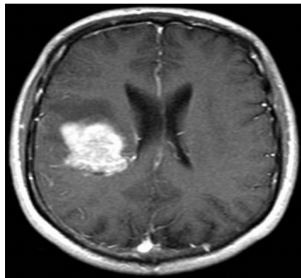
Key Information:

- ▶ Reference for PSNR and MSE
- ▶ Baseline for all comparisons
- ▶ Histogram shows distribution

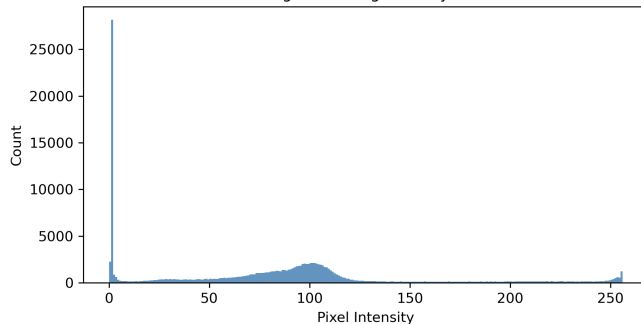
Stage 0: Original Grayscale Image and Histogram

Stage 0 · Original

Original Grayscale Image



Histogram — Original Grayscale



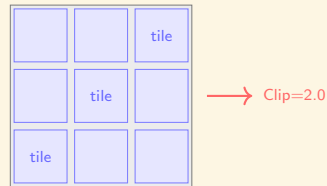
Stage 1: CLAHE Enhancement

Algorithm:

- ▶ Divide image into 8×8 tiles
- ▶ Per-tile histogram equalisation
- ▶ Clipping limit = 2.0
- ▶ Interpolate between tile boundaries

Benefits:

- ▶ Enhanced local contrast
- ▶ Prevents noise amplification
- ▶ Reveals subtle features
- ▶ Maintains natural appearance

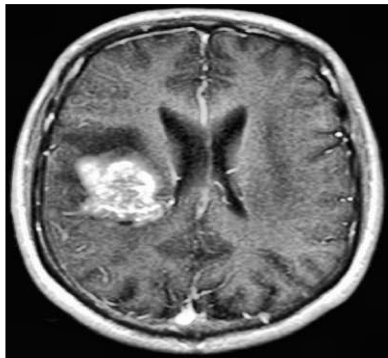


8×8 Adaptive Tiles

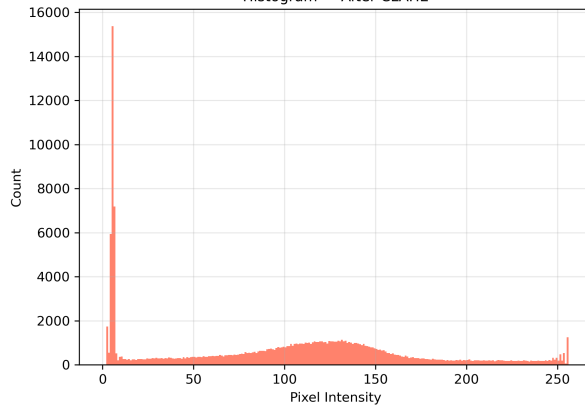
Stage 1: CLAHE Enhancement Result

Stage 1 · CLAHE Enhancement

After CLAHE



Histogram — After CLAHE



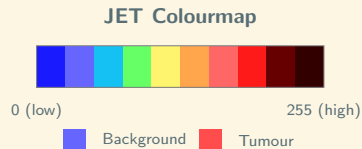
Stage 2: Heatmap Visualisation

JET Colour Mapping:

- ▶ Apply JET colourmap (blue to red)
- ▶ Dark colours: low intensity (background)
- ▶ Bright colours: high intensity (tumours)

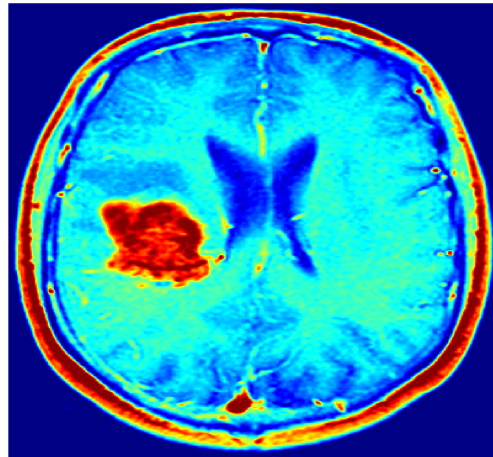
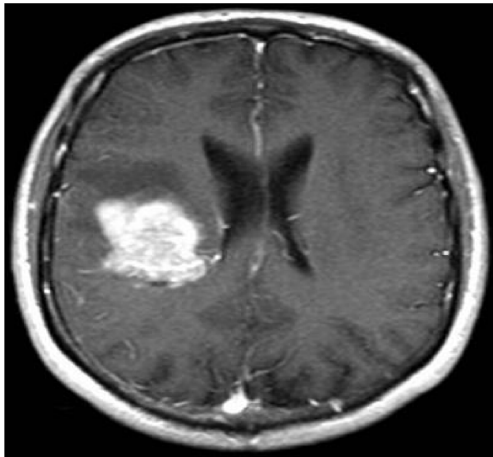
Clinical Application:

- ▶ Rapid hotspot identification
- ▶ Intuitive visual representation
- ▶ Radiologist interpretation aid



Stage 2: Heatmap Visualisation Result

Stage 2 · Heatmap Visualization — Intensity → Color Mapping
Original Grayscale Heatmap (JET colormap)



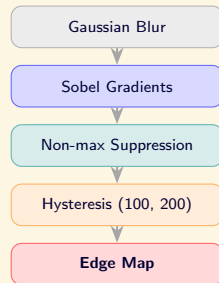
Stage 3: Canny Edge Detection

Algorithm Steps:

- ▶ Gaussian blur for noise reduction
- ▶ Sobel gradient in x and y directions
- ▶ Non-maximum suppression
- ▶ Hysteresis thresholding (100, 200)

Advantages:

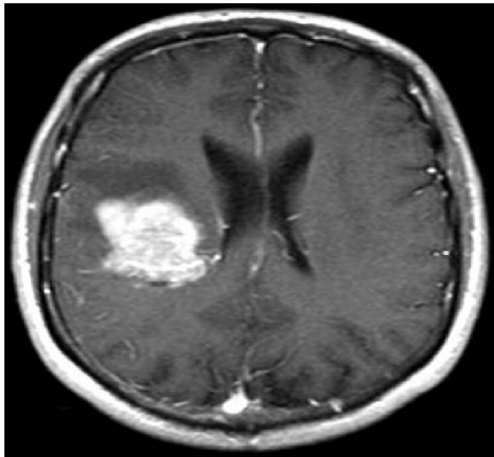
- ▶ Precise structural boundaries
- ▶ Low false-positive rate
- ▶ Connected edge chains
- ▶ Robust to noise



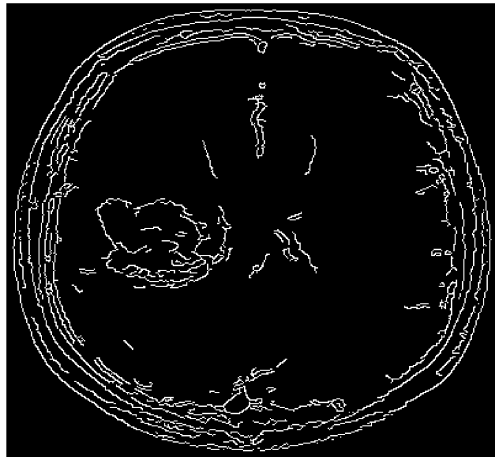
Stage 3: Canny Edge Detection Result

Stage 3 · Canny Edge Detection — Visual Result

Original Grayscale



Canny Edges



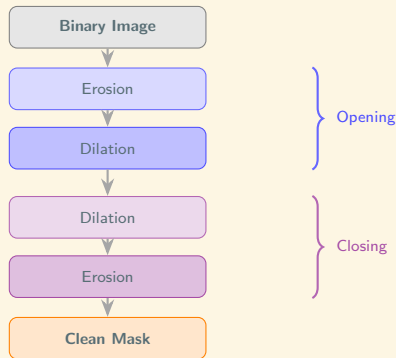
Stage 4: Morphological Operations

Filtering Sequence:

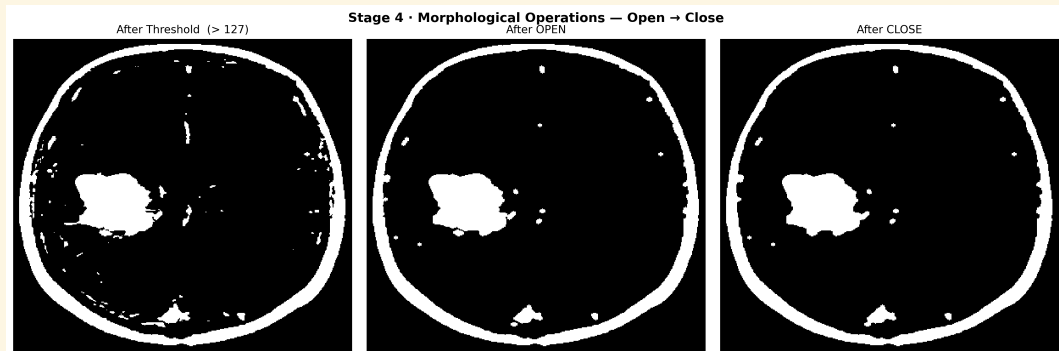
1. Binary threshold (= 127)
2. Opening: Erosion then Dilation
3. Closing: Dilation then Erosion
4. Kernel: 5×5 elliptical element

Effects:

- ▶ Removes small artefacts
- ▶ Fills holes in regions
- ▶ Preserves main structures



Stage 4: Morphological Operations Result



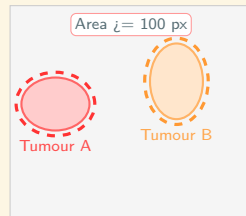
Stage 5: Contour Detection

Connected Component Analysis:

- ▶ Find contours in binary image
- ▶ Filter by area (minimum 100 pixels)
- ▶ Extract boundary coordinates
- ▶ Compute shape properties

Output Features:

- ▶ Tumour regions identified
- ▶ Boundary coordinates extracted
- ▶ Area and perimeter computed
- ▶ Feature vectors for ML



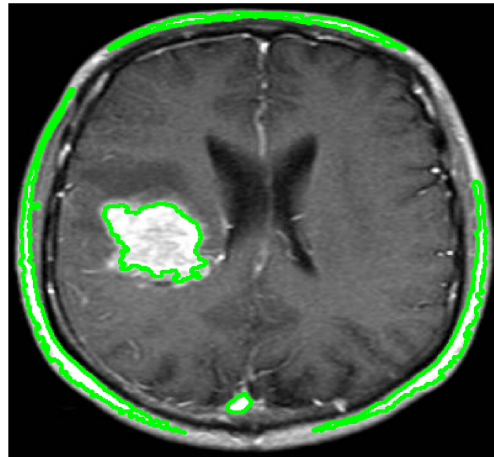
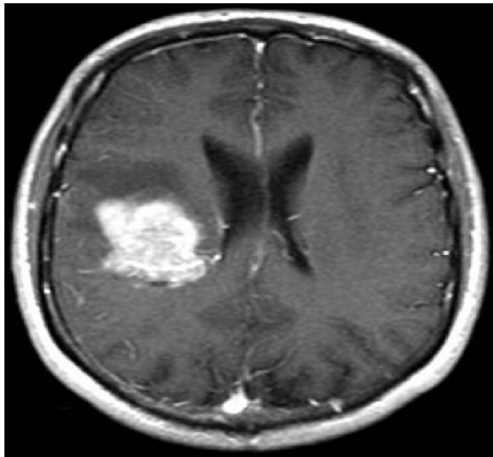
Detected Contours

Stage 5: Contour Detection Result

Stage 5 • Contour Detection — Visual Result

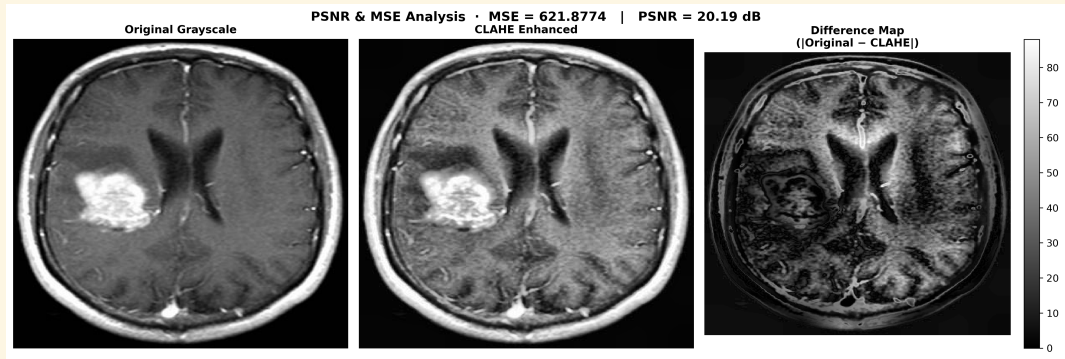
Input Grayscale

Contours Detected (n = 5, min area = 100)



Results

Image Quality Metrics Analysis



Measured Values:

MSE621.88

PSNR20.19 dB

- ▶ Avg. pixel difference ≈ 25 units (0–255)
- ▶ **Expected** for aggressive feature extraction
- ▶ Reflects intentional structural modification

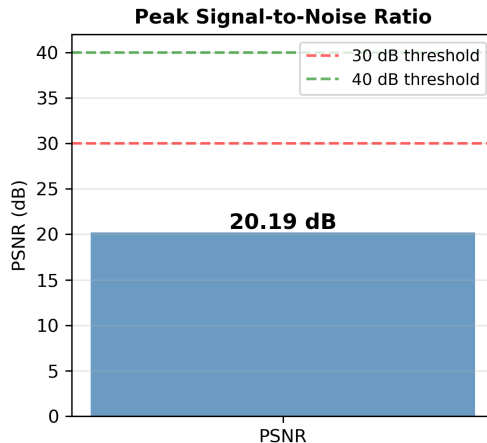
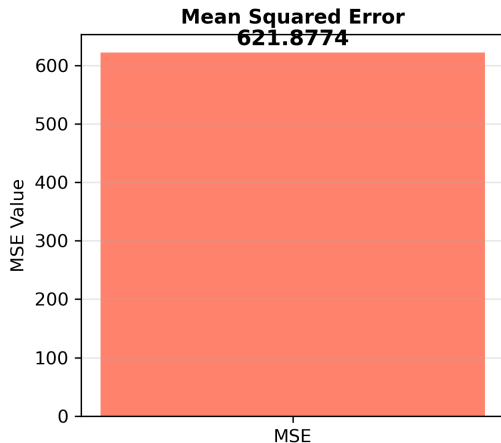
Clinical Context:

- ▶ Goal is feature clarity, not signal fidelity
- ▶ Lower PSNR indicates successful extraction
- ▶ Trade-off: preservation vs. diagnostics



PSNR / MSE Comparative Analysis

PSNR & MSE — Metric Summary (Original vs CLAHE)



Mathematical Foundation

Quality Assessment Equations

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{M \times N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I_{\text{orig}}(i, j) - I_{\text{proc}}(i, j)]^2$$

Peak Signal-to-Noise Ratio (PSNR):

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad \text{where MAX} = 255$$

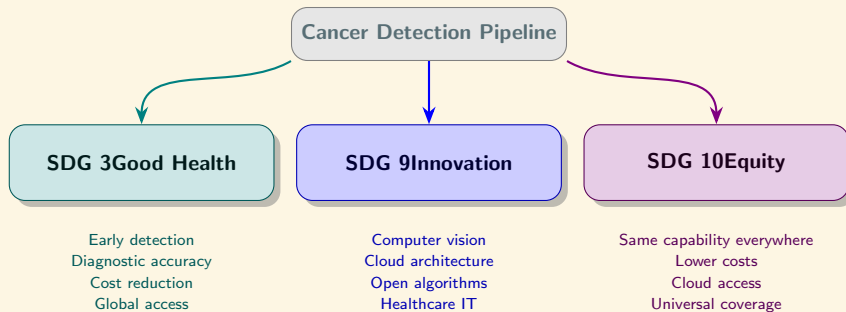
> 30 dB: High Quality

20–30 dB: Acceptable

< 20 dB: High Transform

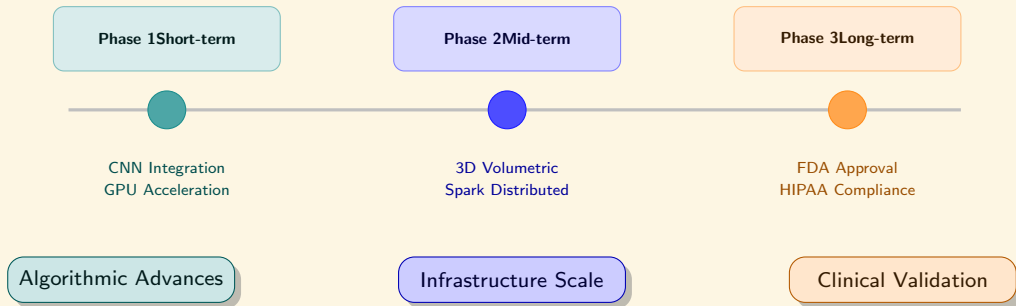
Our result: 20.19 dB

Sustainable Development Goals



Future Enhancements

Future Roadmap



Conclusions

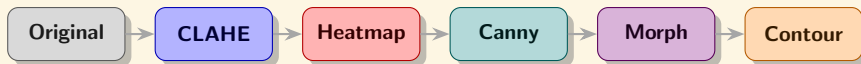
Key Achievements

Technical Success:

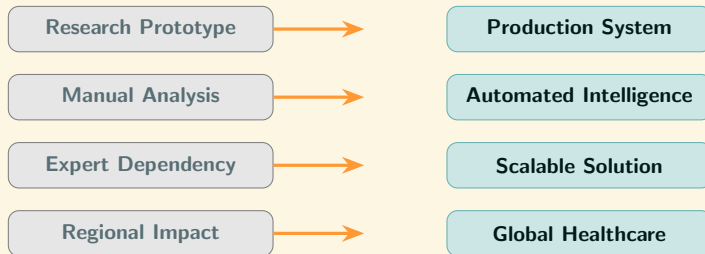
- ▶ 6-stage pipeline fully implemented
- ▶ Multi-feature extraction working
- ▶ Quantitative metrics computed
- ▶ Cloud-native architecture deployed

Clinical Impact:

- ▶ Objective analysis capability
- ▶ Early detection support
- ▶ Accessible technology platform
- ▶ SDG 3, 9 and 10 alignment



From Research to Global Impact



Questions and Discussion

Repository: `github.com/Muhammad-Ahmad17/cancer_cells`

Contact: `muhammadahmad171105@gmail.com`

